

Premonsoon-2025 Inspection Report

Part A - Inspection Details

Salient Features of the Project

Sr.No.	Particulars	Details
1	Dam Name	DEHRANG
2	Dam Type	Earthen dam with gated spillway.
3	Dam Owner	Panvel Municipal Corporation
4	Dam Operator	Panvel Municipal Corporation
5	Commissioning Date	1957
6	State City	Maharashtra
7	District	Raigad
8	Latitude	19° 1' 54"
9	Longitude	73° 14' 32"
10	MWL(m)	89.15 (being automatic gated spillway)
	FRL(m)	89.15
	Spillway_Crest_Elevation (m)	86.65
	MDDL_RL	80.77
	Lowest_River_Bed_Level(m)	75.40
	Deepest_Foundation_Level(m)	NA

Inspection details

Sr.No.	Particulars	Details
1	From Date - To Date	26/04/2025 to 26/04/2025
2	Weather Conditions on Inspection Date	Clear and bright
3	Type of Inspection	Pre- monsoon 2025
4	Reservoir Level (m) on Inspection Date	84.10
5	Inspecting Officers Names and Designation	Raghunath Relekar. Consulting civil engineer. For Yash Engineering Consultant Pvt Ltd. Navi Mumbai
6	Other Remarks	NA

Tree Structure

Part B - Inspection Checklist

Sr.No.	Inspection Item	Response	Response Description	Observation and recommendations, if any, of the authorized inspecting officer
A	Reservoir	Dehrang dam		
A 1.1	General Condition			
1.1.1	Is the reservoir water level unusually high or low?	Very Good	The water level is in commensuration of the use and as per Rule curve.	Normal
1.1.2	Are there signs of decline in water quality?	NO	Water Quality measurement (Either by Dam agencies/ concerned WRD department on the river in the vicinity of Dam) is in place and periodical review of observed water quality is being done. There is no change in water quality	Normal
1.1.3	Are there signs of recent sediment deposition?	Average	No sediment management. However, increasing rate of sediment deposit is observed. At present, this deposition is not affecting any functioning of the dam	Sediment in dead storage is increasing. Silt load in live storage is affecting the utilizable storage.
1.1.4	Is floating debris present?	NO	No floating debris is present.	Floating debris if any expected to wash away in the monsoon flow over crest.
1.1.5	Any indications of major active or inactive landslide area in the reservoir run. If so, indicate their locations and extent.	Good	Heavy landslides are not observed along the periphery of the submergence. Minor slides do not affect safety of the dam and components.	No major issue.
1.1.6	Are there people or livestock in and around reservoir?	Yes	There is habitation at many places along the submergence Periphery, especially farmhouse culture.	No issue in dam safety point of view.
1.1.7	Any other issues?	No		

Sr.No.	Inspection Item	Response	Response Description	Observation and recommendations, if any, of the authorized inspecting officer
B	Dam and Dam Reach (Embankment)	Dehrang dam		
B 1.1	General Condition			
1.1.1	Any major alterations or changes to the dam since the last inspection?	No	Embankment of the dam is in good condition, however slight settlement of the U/S slope. (about 0.35m)	No alterations from last inspection.
1.1.2	Is there any new nearby development in the downstream floodplain?	No	No change in the downstream floodplain development since last inspection	No significant change is observed.
1.1.3	Any misalignment of poles, fencing or walls due to dam movement?	No	No change since last inspection	No
B 1.2	Upstream Slope			
1.2.1	Any signs of bulging or concavity (depressions)? If so, indicate their locations and extent. (Check up the cross-sections with tape and level at random locations, at least two)	Yes	Concavity in the slope with the settlement to the tune of 0.50 m is observed above present water level. The pitching is disturbed.	Embankment is safe.
1.2.2	Does the section of the dam and upstream slope appear structurally sound and stable?	Yes.	No significant sign of distress is observed.	Yes.
1.2.3	Presence of longitudinal or transverse cracks?	No	No cracks in the embankment are seen.	Good
1.2.4	Whether any signs of distress to stability of slopes noticed at any time in any part of the dam? If so, give brief details of the incidents and location, the method of treatment adopted and its effectiveness. Indicate the general condition of upstream pitching.	No	Slopes of the dam are stable enough.	No distress is seen.
1.2.5	Any degradation to slope protection (rip-rap)?	No	The part of the visible pitching shows degradation.	Fair.
1.2.6	Is there any profuse growth of bushes or weeds over any portion of the dam? If so, indicate the locations.	No	The growth of Bushes is cut down timely.	The care is being taken to cut the growth of bushes.
	Does the upstream slope show		No signs up to the open surface	

Sr.No.	Inspection Item	Response	Response Description	Observation and recommendations, if any, of the authorized inspecting officer
1.2.7	existence of crab holes or holes made by rodents or burrowing animals or ant hills? If so, indicate the locations.	No	above water level.	Fairly Good.
1.2.8	Any other issues?	No		No.
B 1.3	Crest of Dam			
1.3.1	Is the crest profile at proper elevation? (To be test checked at random locations by taking level)	Yes	Definitely yes	The crest of dam is fairly at proper elevation.
1.3.2	Does it show any signs of excessive and /or uneven settlement? If so, indicate such locations and extent of settlement. (Surface settlement points must be installed for observing this aspect)	No	No uneven settlement is seen.	Top of the dam is in Good condition.
1.3.3	Is the surface of the crest free from undulations and local depressions or heaving?	Yes	No signs of undulations	Very good top surface.
1.3.4	Does it provide an all- weather road surface?	Yes	Definitely yes	Yes
1.3.5	Any degradation to access road (sealed /unsealed)?	No	Unsealed road surface provides all weather road up to NOF part.	Good road available up to left side flank wall. The OF part of the dam is not approachable for movement of machinery in the case of emergency
1.3.6	Does it develop any visible cracks in transverse or longitudinal directions? If so, attach a map showing their locations and extent. Depth of cracks must be ascertained by taking open trenches extending below the bottom of cracks.	No	No significant cracks are seen,	Fair road condition.
1.3.7	What is the condition of the edges of crest? Have got eroded and cut up resulting in	No	Definitely no	Road width available is fairly as per the design.

Sr.No.	Inspection Item	Response	Response Description	Observation and recommendations, if any, of the authorized inspecting officer
	reduced effective width?			
1.3.8	Is the crest free from local slips throughout its length on either side?	Yes	Crest is free from local slips.	Fairly good.
1.3.9	Do the headers, guard stones and parapet wall provided at the edges of the crest appear in proper profile and plumb?	Fair	Fairly Yes.	Guard stones are available in proper condition.
1.3.10	Any degradation to upstream parapet or downstream curb wall?	NA	No provision	NA
1.3.11	Evidence of livestock on dam crest?	No	No major live stocks are on the dam crest.	Fairly no
1.3.12	Trees or profuse growth of weeds /bushes at any location?	No	No.	Presently efforts are taken by the dept. to cut the vegetation.
1.3.13	Proper lighting arrangement at dam top?	Yes	Good lighting arrangement with Solar power is installed on the dam top.	Being the day time. The performance couldn't be observed. They are functioning as reported by the dept.
1.3.14	Any other issues?	Yes	Good approach of all-weather road is necessary for OF section of 111m long.	Needs action to provide the road or bridge for right flank
B 1.4 Downstream Slope				
1.4.1	Any signs of bulging or concavity (depressions)?	No	No signs	No bulging or concavity is observed

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1.4.2	Are there any wet or slushy patches or any concentrated leaks, springs or trickles observed on the downstream slopes or the toe? If so, indicate their locations and extent. Please look out for patches of extensive vegetation growth and examine them carefully and record the findings	No	No signs	No such case of the wet or slushy surface. Vegetation growth with shrubs and small trees are being removed by the department.
1.4.3	Presence of longitudinal or transverse cracks?	No	No signs	No Significant Cracks are observed.
1.4.4	Any signs of distress to the stability of slopes?	No	No signs	No
1.4.5	Are rain cuts/erosion channels present at any location?	Yes	Rain cut at the toe and left side abutment of the dam and erosion on the D/S of the dam at about - 20m from the left end of the dam is seen.	The erosion needs immediate remedy. Efforts are being made to fill up the Rain cut
1.4.6	Are all the rain cuts and erosion channels properly treated and made good? Please indicate location of recurring damages, if any.	Yes	No prominent rain cuts are seen on the D/S slope.	No
1.4.7	Is there any profuse growth of bushes or weeds over any portion of the dam? If so, indicate the locations.	Average	Grass and shrubs are grown at the D/S toe of the dam in almost full length of the dam.	Shrubs and grass Growth is Cleared Periodically.
1.4.8	Does the downstream slope show existence of crab holes or holes made by rodents or burrowing animals or ant hills? If so, indicate the locations.	No	No signs	No holes or cavities are seen on the D/S slope.
1.4.9	Any other degradation to slope protection (turfing)? Indicate the general condition of downstream pitching/ turfing and rock toe.	No	In general, the D/S slope protection is intact.	The D/S protection is in good condition.

Sr.No.	Inspection Item	Response	Response Description	Observation and recommendations, if any, of the authorized inspecting officer
1.4.10	Is the downstream area clear of debris and free draining?	No	The Area near D/S toe is covered with the vegetation.	No drains are functioning, and the open part of the C and Toe drain is partly silted and is being cleared.
1.4.11	Any other issues?	No		
B 1.5 Downstream Drainage				
1.5.1	Are there any signs of water logging, slushy conditions or growth of aquatic weeds on the downstream of the dam? To be checked up to 300m downstream of toe	No	No water logging is observed beyond the D/S toe of the dam	No such case.
1.5.2	Are there any standing pools of water in the downstream of dam? If so, give their locations and extent. To be checked up to 300m downstream of toe	Very Good	No standing pool of water is observed beyond the D/S toe of the dam	No such case
1.5.3	Are there any boils observed in the vicinity of the downstream toe of the dam? If so, give locations.	Very Good	No water boils are observed beyond the D/S toe of the dam and in the vicinity	No such case
1.5.4	Is the downstream area sufficiently clear and free draining?	Yes	Vegetation growth is seen at the toe of the dam.	The vegetation growth is being cleared by the dept.
1.5.5	What is the depth of ground water table on the downstream as evident from the existing wells in the vicinity of the dam to be checked up to 300 m downstream of toe. Does the water table show any marked	NA	The ground water table could not be ascertained as there is no well at ground level below the toe of the dam.	No such case is observed.

Sr.No.	Inspection Item	Response	Response Description	Observation and recommendations, if any, of the authorized inspecting officer
	variation in accordance with the variations in reservoir water level?			
1.5.6	Are all the exposed drains working satisfactorily?	No	No exposed part of C and toe drains are functioning	No drains are functioning
1.5.7	Toe drains and cross drains. I. Are the portions of longitudinal toe drain and exposed cross drains beyond the downstream toe of the dam in regular section and freely draining? II. Is the pitching to these drains intact? III. Is there any weed growth in these drains? IV. Indicate other defects noticed in the drains, if any.	Yes	Toe drains and cross drains are in regular section and freely draining. Pitching of these drains are intact. Weed growth in the drains is noticed and being removed	Though the section of Toe drains and part of C drain is intact, these are not functioning.
1.5.8	Outfall Drain: (a) Is the outfall drain in proper shape and grade and freely draining? (b) Is the outfall drain properly cleaned and maintained? Does the outfall drain show any stagnant pools of water or weed growth?	Yes	NO stagnant pools are observed at Outfall drain	Structurally ok but not functioning
B 1.6	Surface Drainage of Downstream Slope			
1.6.1	Is the condition of the downstream slope drainage arrangements, if provided, satisfactory?	Yes	Longitudinal surface drain on the first berm is provided	Fair in condition.
1.6.2	Is the paving to these drains intact?	Yes	Fairly intact.	Yes
1.6.3	Are all the drains properly maintained and free of vegetation growth and debris?	Yes	Fairly maintained.	Yes
1.6.4	Does the slope have a tendency to develop severe rain cuts at any location?	Average	Somewhat	Yes
1.6.5	Enumerate any other defects noticed in the surface drainage of downstream slope.	No	Not seen	No

Sr.No.	Inspection Item	Response	Response Description	Observation and recommendations, if any, of the authorized inspecting officer
B 1.7	Seepage Measurement			
1.7.1	Is the quantity of seepage being daily or periodically measured and recorded? Please check the registers and record observations.	NA	NO seepage from the dam body and hence No provision,	No provision
1.7.2	Does it show any abnormal rise or fall? If so, does it have any relation to a certain reservoir level elevation?	NA		--
1.7.3	Does the seepage show a turbid colour at any stage? Was such a phenomenon observed at any stage at any location in the past?	NA	No seepage is observed in past.	No
1.7.4	What is the measured rate of seepage flow with date and reservoir level; 1. On the day of present inspection 11. Maximum since last June Ill. Minimum since last June	NA		--
1.7.5	Is the portion upstream and downstream of measuring points of seepage easily accessible with proper steps and paths and free of vegetation growth?	NA		--
1.7.6	Are the measuring points properly located, constructed and maintained so as to give accurate and reliable measurements of seepage in accordance with the relevant IS Codes?	NA		--
1.7.7	Is the method of taking seepage measurements satisfactory?	NA		--
B 1.8	Breaching Section (if Provided)			
1.8.1	Is the breaching section easily accessible?	NA	No provision	No provision

Sr.No.	Inspection Item	Response	Response Description	Observation and recommendations, if any, of the authorized inspecting officer
1.8.2	Is the condition of the breaching section satisfactory?	NA		--
1.8.3	Is the note of instructions as to when and how to operate the breaching section available on record?	NA		--
1.8.4	For reconstruction after the breach are the following items decided in advance? a) Quarry for embankment material b) Suitable routes of access Is the maintenance staff fully aware of the instructions related to operation of the beaching section and for reconstruction after the breach	NA		--
1.8.5	Ascertain and indicate the latest event of operation of breaching section and its performance.			
1.8.6	Evidence of recent degradation?			
1.8.7	Any other issues?			
B 1.9	Junction of Earth work with Masonry/Concrete dam sections and outlets			
1.9.1	Is there any existence of leaks, springs or wet spots in the earth work in the vicinity of the junctions between earth work and masonry works? If so, what is the approximate rate and colour of the leakage? Does it turn turbid at any time? Please ascertain from enquiries and record the findings.	No	No signs of leakages are observed.	No leakages occurred in past as reported by the site in charge.
	Is there any tendency for separations, cracking, settlement or upheaval of the	No	No such signs are observed.	No such case.

Sr.No.	Inspection Item	Response	Response Description	Observation and recommendations, if any, of the authorized inspecting officer
1.9.2	earth work in the vicinity of masonry or concrete? If so, indicate the locations and the exact nature of deficiency.			
1.9.3	Is there any tendency for surface erosion or slope instability at the junction?	No	No such signs are observed.	No such case
1.9.4	If the outlet conduit is located in the earth dam section, is the entire length of the conduit in perfect order and profile and free from offsets, open joints, cracks and leakage? Examine the conduit carefully from the downstream or from inside, if possible, and indicate the deficiencies observed, if any.	Yes	This is being a Water supply dam, the same provided with multiple rows of C I pipe outlets of diameter less than 30cm . These conduits are above the Sill and at different levels. The conduits cannot be inspected from inside.	There is no sign of any disorder in the whole conduit length.
1.9.5	Any other issues?	No	No	
B	Relief Wells			
1.10				
1.10.1	Are the relief wells in good working condition and functioning well?	NA	No provision	No provision
1.10.2	Are the relief properly surged and cleaned periodically?	NA		--
1.10.3	Please indicate the dates of last surging and cleaning and the next surging due.	NA		--
1.10.4	Are the necessary plant and equipment for cleaning the relief wells, available with the office?	NA		--
1.10.5	Is the record of periodical measurements of discharge from each relief well maintained? If so, indicate total discharge and maximum discharge observed from a single well on the date of inspection.			

Sr.No.	Inspection Item	Response	Response Description	Observation and recommendations, if any, of the authorized inspecting officer
B	Abutment Contacts			
1.11				
1.11.1	Any presence of leaks, springs or wet spots near the abutment?	No	No signs	No such case
1.11.2	Any presence of cracking, settlement or upheaval of earthwork?	No	No signs	No such case
1.11.3	Any evidence of erosion or slope instability?	No	No signs	No such case
1.11.4	Trees or profuse growth of weeds /bushes?	No	No signs	No such case
1.11.5	Any degradation to up /downstream slope protection (rip-rap, turfing)?	No	No signs	No such case
1.11.6	Any other issues?	NA		
E	Spillway and Energy Dissipation Structure	Dehrang		
E 1.1	Spillway			
1.1.1	Any problems inspecting spillway (obstructed access, damaged catwalk)?	No	No obstructions in inspection.	Catwalk bridge is needs corrosion proof painting
1.1.2	Any obstructions in or immediately downstream of the spillway?	No	No potential for blockage of sluiceways or outlets by debris	No obstructions on D/S of the spillway.
1.1.3	Evidence of abrasion, cavitation or scour on glacis (e.g. exposed reinforcement)?	No	No evidence of damaged on the glacis.	Glacis surface is intact.
1.1.4	Presence of displaced, offset or open joints?	No	Not observed.	No such case.
1.1.5	Presence of cracking (structural, thermal, along joints)?	No	No joints are exposed.	No such case.
1.1.6	Evidence of surface defects (honeycombing, staining, stratification)?	No	No evidence. Well maintained Cement concrete surface without Honey combing.	No such case.
1.1.7	Concrete /masonry deterioration (spalling, leaching, disintegration)?	Yes	No such case with concrete surface. Masonry of flank wall is damaged at bottom.	As observed.

Sr.No.	Inspection Item	Response	Response Description	Observation and recommendations, if any, of the authorized inspecting officer
1.1.8	Presence of vegetation (growth in joints between blocks)?	Yes	There is growth of vegetation on the surface of flank wall masonry. Leakages at bottom of FW inside the bucket is seen	Earlier growth is cut down and new growth at few places is seen
1.1.9	Evidence of any other damage to joints and /or water stops?	No	No damages to joints. Water stops can't be located.	As observed.
1.1.10	Excessive seepage /sweating at any location on spillway glacis?	No	No seepage or sweating is present on the spillway glacis	No such case.
1.1.11	Significant leakage at any location on spillway glacis?	No	Present water level is much below the crest.	As observed
1.1.12	Any other issues?	Yes	Automatic Godbole type gates 11 nos are provided for surplussing arrangement. The hinges are required good quality oiling and interim maintenance	The gates provided on the crest tilt with water pressure of flood. However, it requires additional mechanical force for closing or opening .
F	Intake/Outlet and Water Conveyance Structure	Dehrang		
F 1.1	Intake Outlet Structure			
1.1.1	Any problems inspecting intake /outlet structure (obstructed /unsafe access)?	Yes	Intake for water supply is provided in small diameter pipes and Top-level outlet in exposed and is above present water level. There is no obstruction in three outlet pipes provided at different levels. Scouring pipe of 600mm dia. is provided below service pipes and is silted.	No problem in service pipes. A scour (pipe) outlet is choked up with silt and needs cleaning
1.1.2	Any obstructions in, upstream or downstream of intake /outlet structure?	No	No obstructions observed over the years in service outlets. Well maintained/ Standard practices are being followed all the time.	Obstructions observed only in scour pipe.
1.1.3	Evidence of abrasion, cavitation or scour on intake /outlet structure?	No	No evidence. Well maintained and being operated. Standard practices are being followed all the times. Outlet pipes are not visible.	No such case.

Sr.No.	Inspection Item	Response	Response Description	Observation and recommendations, if any, of the authorized inspecting officer
1.1.4	Any evidence of structural distress (displaced/offset/open joints, cracking)?	No	System seems to be in good condition. There is no leakage from the body of the dam at the section of outlet on D/S side.	The evidence of no leakages infers the pipes are intact.
1.1.5	Any evidence of surface defects and/or concrete /masonry deterioration?	No	No evidence. Well maintained/ Standard practices are being followed all the time.	Not visible and accessible.
1.1.6	Any other issues?	Yes	Scour pipe needs cleaning and washing away the silt. This will lead to increase in the live storage.	Cleaning of the scour outlet is necessary in Emergency point of view. Efforts are being made to clean the pipe.
H	Access Road	Dehrang		
H 1.1	General Condition			
1.1.1	Any problems ensuring security of dam site (including gates and fencing)?	Yes	Good security with gate structure is available.	Yes
1.1.2	Is there a properly constructed and well-maintained all-weather access road to the dam site?	Yes	Good Access road is available up to dam site. The dam top provides approach up to the flank wall of OF. Section.	The MDR asphalt road up to dam site and good WBM road up to dam is available.
1.1.3	What is the type of the pavement of the access road and its condition?	Very Good	Well maintained MDR tar road followed by the WBM road is available and is well maintained throughout the year.	Asphalt Road followed by the WBM road
1.1.4	Are there properly constructed and well-maintained access road arrangements to the following components for inspection, maintenance and repairs? Top of Dam Spillway Gates and hoisting arrangement Drainage gallery, adits and exist Bridge structure Downstream stilling basin Junction and abutments Outlet control tower Outlet gates Toe of earth dam, downstream drainage arrangements and berms. All saddle dams.	Partly Yes	Good Access road is available up to dam site. The dam top provides approach up to the flank wall. The OF section of total length 111m with 11 nos gates have no approach for machinery in case of emergency.	Partly yes

Sr.No.	Inspection Item	Response	Response Description	Observation and recommendations, if any, of the authorized inspecting officer
1.1.5	What is the general condition of all the masonry structures on various access roads?	Good	Good	NA
1.1.6	Are all the structures on the access roads adequately safe for allowing passage of plant machinery for emergent repairs?	Yes	Good enough to carry the construction machinery in case of emergency.	Fairly adequate road access is available.
1.1.7	Any obstructions along or at entrance to access road (temporary or long- term)?	No	No major obstructions are there.	No
1.1.8	Any slope stability issues (road embankment or adjacent slopes)?	No	No issue of the slope stability	No
1.1.9	Profuse growth of weeds /grass on or in vicinity of access road?	No	Regular treatment is being done to remove weeds/grass on or in vicinity of access road	No weeds growth is observed.
1.1.10	Any drainage problems (standing water on or adjacent to road)?	No	Not seen.	No such case.
1.1.12	Any other issues?	NA		
K	Emergency Preparedness	Dehrang		
K 1.1	Emergency Action Plan			
1.1.1	Is the Emergency Action Plan (EAP) prepared for the dam as per the national guidelines? If not, the expected date of preparation of guidelines	Average	Existing EAP is being updated as per the regulations of Central Govt.	Need to be revised.
1.1.2	When EAP was last updated?	Average	within one to five years	NA
1.1.3	"If not, are any dam staff unaware or insufficiently conversant with the EAP?"	Average	Dam staff have attended training session or participated in a simulated drill in past, but not within one year	NA
1.1.4	"Any concerned authorities unaware or insufficiently conversant with the EAP?"	Average	Concerned authorities have not participated in training sessions/simulated drill but attended stakeholder meeting before one year	NA

Sr.No.	Inspection Item	Response	Response Description	Observation and recommendations, if any, of the authorized inspecting officer
1.1.5	Are communication directories /contact details and other dynamic information are being updated annually?	Yes	Yes. The contacts are being updated.	Yes.
1.1.6	Any problems accessing or operating the communication / warning system?	No	Problem has been identified and are being resolved shortly.	NA
1.1.7	Are inundation maps updated and available to concerned authorities?	No	Inundation map prepared/updated by basic level screening and simple analysis using low resolution terrain data (SRTM, ASTER, or 30 m INTERMAP)	The updating is under process
1.1.8	Are the concerned authorities informed about the system of emergency reporting procedures and warning?	Average	EAP has been distributed to all concerned authorities. Stakeholders' annual consultation meeting will be held before monsoon.	Yes
1.1.9	Are available safety spots on the downstream of the dam identified and made known to the concerned authorities? Are adequate warning devices and facilities provided at the dam?	Yes	Partial arrangement of warning devices/facilities, and equipment and supplies have put in place for emergency situations	Yes
1.1.10	Are proper arrangements made for security of the dam and preventing cases of unauthorized trespass, vandalism and sabotage to the dam works?	Yes	Complete security arrangements have been ensured such as deployments of guards, CCTV surveillance, gates and wire fencing etc.	Yes
1.1.11	Date of last annual stakeholder consultation meeting along with mock drill exercise conducted	Average	Annual stakeholder consultation meeting/mock drill exercise are not being conducted regularly. <i>Efforts will be made to conduct meeting of stakeholders before onset of the monsoon</i>	No
1.1.12	Does the EAP disseminated to all the concerned stakeholders?	No		

Sr.No.	Inspection Item	Response	Response Description	Observation and recommendations, if any, of the authorized inspecting officer
1.1.13	Any other issues?	No		
K 1.2 Inspection of Records				
1.2.1	Whether following Dam Safety Documents are prepared and approved by the competent authority? I. As Built Drawings II. EAP III. Completion Report IV. Data Book V. O and M manual	Average	<i>All old record is not available. The efforts are being made to prepare the documents as on site.</i>	No
1.2.2	Are the relevant documents reviewed and updated from time to time?	Average	More than one but less than three documents mentioned at point 1.2.1 have been reviewed and updated as per the regulations of Central Govt.	No
1.2.3	Are all the members of the maintenance staff adequately trained and fully conversant with their responsibilities concerning. (a) Designer's Operation Criteria. (b) Standing Operating Procedures. (c) Maintenance and Vigilance Procedures of the dam (d) Maintenance and operation of all control equipment. (e) Reservoir Operation Schedules, Gate Operation Schedule (f) Maintenance and Operation of all instruments. (g) Identification of signs of deficient behavior. (h) Reporting Procedures of emergency situations. (i) Emergency repairs.	Average	This dam is having very small storage, in Emergency Action point of view. The staff in charge of the dam is trained adequately. However, the updating of the Dam Break Analysis, revision of the EAP, O and M manual etc. is taken in hand.	Yes
<u>Health Condition of Component</u>				
Category III - Minor deficiencies which require rectification during the year.				

Part 2B – Consolidated Dam Health Status Report (Dehrang)

Sr. No.	Observation / Signification Deficiencies Noticed	Remedial Measure Suggested
1	Open cross drain and Longitudinal Toe Drains Are Found Silted. All the drains inside the body of dam may not be functioning.	Open C and Toe drains <i>are being</i> Cleaned for Functioning. Drains in the body of dam can't be repaired.
2	Scour sluice should be cleaned for the depletion of the dam in case of emergency.	The silted pipe and non-operating of the valve of the scour sluice needs to be cleaned with the help of Divers if needed.
3	The valve operating (stem) rods are not standard. These should be changed. The MS cover over chamber is badly rusted.	This needs total replacement of rods, turning mechanism and chequered plate over the chamber.
4	Head regulator well and inlet chamber over it shows leakage. The source of leakage could not be ascertained due to lack of light.	The leakages should be stopped to keep the well dry.
5	Big erosion on the D/S toe near starting point of dam should be filled and protected. This reduces the support of the dam body.	The big pit should be filled with soil, compacted with slope protection measures.
6	The approach road for overflow section is not available to carry machinery during emergency.	Alternate road or bridge must be constructed on the Overflow section in the vicinity of the spillway structure. <i>The same is being studied.</i>
7	The level measuring gauge plates for reservoir are damaged and needs replacement. Outflow measuring gauge plates on the divide wall, and on flank wall in both the bays of EDA needs to be provided.	Needs immediate action
8- Second priority	The U/S embankment section needs to be rectified as per the designed section with repairs and rearrangement of the pitching.	The dam embankment may be repaired with additional bank work and repairs to the pitching with partial fresh stones.